TURNKEY ROOFTOP SOLAR EPC & SREDA NET-METERING

Boardroom Strategic Decarbonization & 25-Year Energy Cost Insulation Pitchbook

CLASSIFICATION: Strictly Private & Confidential | Boardroom Submittal	**ENTITY:** Ariel Renewable Energy Limited (ArielMC Group)
TARGET AUDIENCE: Board of Directors / Chairman / MD / CEO	**PRODUCT / SECTOR:** Solar Infrastructure / Energy
FINANCING: Bangladesh Bank 4.5% Green Refinancing Scheme	**DATE:** November 2026 | energy.arielmc.com | crm.arielmc.com



SOVEREIGN INDUSTRIAL HERITAGE MEETS CLEAN ENERGY

Executive Address & 25-Year Value Proposition from ArielMC Leadership

• **The Rooftop Goldmine:** Vacant factory shed sheets or commercial slabs can generate clean daytime electricity at an amortized LCOE of **Tk 3.20/unit** (vs. Tk 11.50+ grid).

• **CBAM & ESG Protection:** Protects export orders against European Union Carbon Border Adjustment Mechanism (CBAM) carbon penalties and achieves LEED Platinum points.

• **100% Turnkey Risk Elimination:** Ariel Renewable Energy manages 100% of civil engineering, SREDA net-metering approvals, utility substation sync, and 25-year O&M.;

• **Guaranteed Core SLAs:** 15-Year 100% Water-Tight Guarantee, 180 km/h BNBC 2020 Cyclone Wind Load, and Contractual Performance Ratio PR ≥ 80%.



THE BANGLADESH ENERGY PARADIGM: GRID ESCALATION VS. SOLAR LCOE

Comparative Analysis: Legacy Grid Power, Captive Diesel & Ariel Turnkey Solar

REB / DESCO Industrial (LT-E / HT-3)	Tk 10.80 – 12.20 / unit	+8% to +12% / Year	Fossil-Heavy; Scope 2 Penalty
Commercial Peak Grid (Gulshan/Motijheel)	Tk 13.80 – 15.20 / unit	+10% to +14% / Year	Highest Tariff; Zero Green Points
Captive Diesel Generator (Backup)	Tk 32.00 – 38.00 / unit	Tied to global oil volatility	Severe carbon emission & noise
Ariel Turnkey Solar (SREDA)	**Tk 3.20 / unit (LCOE)**	**FIXED for 25 Years (Tk 0 Fuel)**	**100% Green; CBAM & LEED Certified**

THE SREDA NET-METERING ENGINE: Every daytime exported unit offsets evening/weekend peak loads on a 1:1 bi-directional billing credit, eliminating energy inflation risk for 25 years.

HOW SREDA NET-METERING POWERS YOUR BALANCE SHEET

Daytime Self-Consumption + Automatic Grid Export + Indefinite Monetary Credit Rollover

1. Daytime Generation (9 AM – 3:30 PM): Tier-1 bifacial modules harvest peak solar daylight to feed operating machinery.

2. Zero-Loss Local Consumption: Internal plant loads consume solar power first, reducing grid draw to near zero.

3. Automatic Surplus Export: Weekend or light load surplus exports automatically into the national grid via bi-directional meter.

4. 1:1 Tariff Credit on Bill: REB / DESCO credits every exported kWh against peak evening consumption.

5. Indefinite Credit Rollover: Excess credits roll over continuously across monthly billing cycles.



ENGINEERED PACKAGES: CATEGORY 1 — C&I; FACTORY ROOFTOP

Heavy Industrial, Export RMG, Textiles, Leather Tannery & Jute Mills

F1: Industrial Base-Load	300 kWp	~24,000 sq ft	36,000 – 39,000 kWh	Tk 1.56 Cr (k)	3.2 Years
F2: Megawatt Flagship	1,000 kWp (1.0 MWp)	~78,000 sq ft	125,000 – 135,000 kWh	Tk 4.90 Cr (k)	3.0 Years
F3: Heavy Industrial Complex	2,500 kWp (2.5 MWp)	~200,000+ sq ft	315,000 – 340,000 kWh	Tk 11.87 Cr (k)	2.85 Years



COMMERCIAL BUILDINGS & RESIDENTIAL REAL ESTATE PACKAGES

Office Towers, Shopping Malls, Hospitals, Private Villas & Apartment Societies

Package Code	Capacity	Footprint	Turnkey CapEx (BDT)	Turnkey CapEx (USD)	Payback
C1: Corporate Office Tower	75 kWp	~6,000 sq ft RCC	Tk 46,50,000	,750	3.1 Years
C2: Retail Mall & Hospital	200 kWp	~16,000 sq ft	Tk 1,16,00,000	,600	2.9 Years
H1: Urban Villa Micro-Array	10 kWp Grid-Tied	~800 sq ft RCC	Tk 7,50,000	,250	4.2 Years
H2: Luxury Hybrid + 15kWh LFP	20 kWp + Batt	~1,600 sq ft RCC	Tk 22,00,000	,300	5.5 Years
H3: Apartment Society Common	40 kWp Comm-Grid	~3,200 sq ft Roof	Tk 27,20,000	,600	3.8 Years





TECHNICAL ARCHITECTURE & TIER-1 BLOOMBERGNEF HARDWARE

Global Quality Standards: N-Type TOPCon Bifacial, Multi-MPPT Inverters & BNBC Structures

PV Solar Modules	Jinko Tiger Neo 620W N-Type TOPCon Bifacial Dual-Glass (22.8% eff)	LONGi Hi-MO X6 / Trina 585W Monocrystalline N-Type (21.8% eff)	12-Yr Product | 30-Yr Linear Performance (≥87.4% at Yr 30) | IEC 61215/61730
Grid Inverters	Huawei SUN2000 Multi-MPPT with AI AFCI Arc Fault Extinguishing in 2.5s	Sungrow SG Series C&I; Multi-MPPT (98.6% eff, IP66/C5 corrosion)	5-10 Years Manufacturer Warranty | VDE-AR-N 4105, EN 50549-1, IEEE 1547
Mounting Structures	Aluminium 6005-T5 Anodized with Non-Penetrative Standing Seam Clamps	Hot-Dip Galvanized Steel (HDG) 80-100 micron with SS304/316 Fasteners	15-Year Structural Integrity | BNBC 2020 Cyclone Certified (180 km/h wind gusts)
Switchgear & SCADA	ABB / Schneider Electric Class-A DC/AC SPDs + Motorized MCCBs	Dehn / Chint Industrial Combiner IP65 + Class 0.2S Net-Meter	25-Year Cloud Monitoring License on Ariel-SaaS™ Telemetry Console

TURNKEY INVESTMENT COSTING & 25-YEAR FINANCIAL YIELD

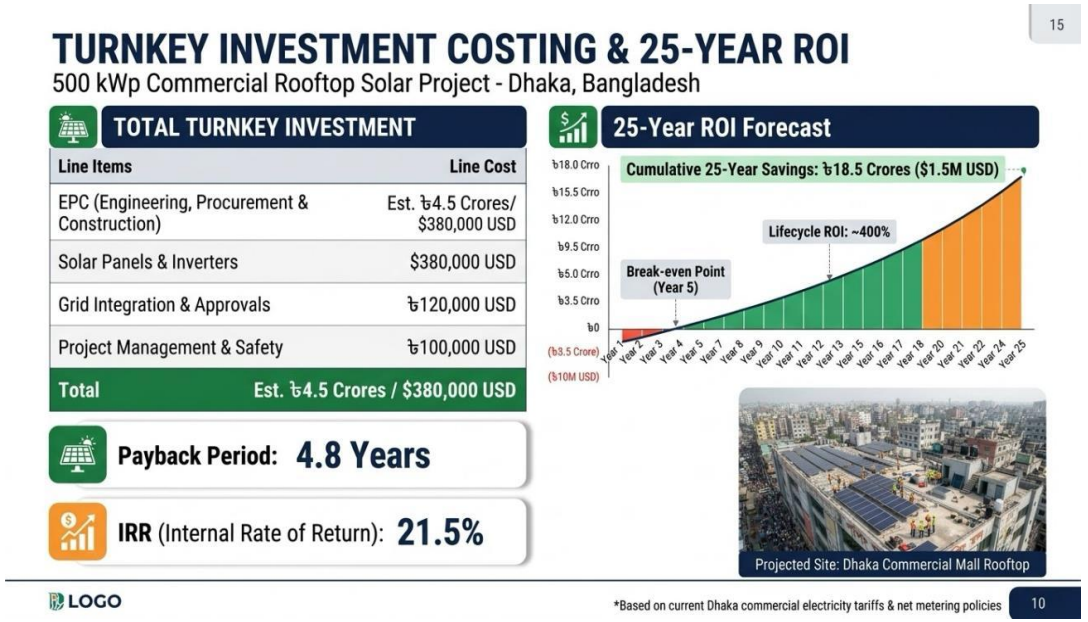
Detailed BOQ Breakdown, Financial Model & 500 kWp Commercial Case Study

Turnkey CapEx Benchmarks:

- 300 kWp Factory (F1): Tk 1.56 Cr (,000) — 3.2 Yrs Payback
- 1.0 MWp Flagship (F2): Tk 4.90 Cr (,300) — 3.0 Yrs Payback
- 2.5 MWp Multi-Roof (F3): Tk 11.87 Cr (,500) — 2.85 Yrs Payback
- 75 kWp Office Tower (C1): Tk 46.50 Lakh (,750) — 3.1 Yrs Payback
- 200 kWp Mall/Hospital (C2): Tk 1.16 Cr (,600) — 2.9 Yrs Payback

Detailed BOQ Capital Allocation:

43% Modules | 18% Inverters/Switchgear | 12% Structures | 8% DC/AC Cables | 6% Substation Sync | 7% Civil Rigging | 6% SREDA Licensing



ZERO WORKING CAPITAL STRAIN: BANGLADESH BANK 4.5% GREEN DEBT

Green Transformation Fund (GTF) & SREDA Solar Window Financing Structure

80% CONCESSIONARY DEBT	POSITIVE NET CASHFLOW	ACCELERATED TAX SHELTER
Borrow up to 80% of project CapEx at 4.5% to 5.0% interest rate over 5 to 7 years.	Monthly electricity savings exceed monthly loan repayments, delivering positive cashflow from Month 1.	Solar CapEx qualifies for rapid corporate depreciation, reducing annual corporate income tax.

FINANCIAL ARCHITECTURE SUMMARY: Client contributes only 20% equity; the remaining 80% is self-liquidated by clean electricity savings without draining operational working capital.

STRUCTURAL SAFETY & RISK ELIMINATION PROTOCOLS

Zero-Penetration Roof Waterproofing, 180 km/h Cyclone Wind Code & AI Fire Arc Defense

Roof Water Leakage	100% non-penetrative aluminium seam clamps for standing seam roofs—zero sheet drilling. EPDM gaskets with Sikaflex UV sealant on trapezoidal metal.	**15-Year 100% Water-Tight Guarantee** backed by written EPC liability.
180 km/h Cyclonic Winds	Engineered strictly in compliance with Bangladesh National Building Code (BNBC 2020) wind zone standards with aerodynamic deflectors.	Certified Structural Engineer (BUET Civil) vetting stamp on every project.
Fire & DC Electrical Arcs	Inverters feature AI-driven AFCI that detects and extinguishes DC electrical arcs within 2.5 seconds (NFPA 855 / IEC 62109).	Zero-fire hazard certified switchgear & halogen-free DC cabling.
Lightning & Surges	Dedicated copper Franklin interception rods and chemical earth boreholes achieve resistance < 1.0 Ohm, isolating factory machinery.	IEC 62305 lightning protection certification.

25-YEAR OPERATIONS, MAINTENANCE & PERFORMANCE SLA

Maximizing Asset Lifetime Yield, Robotic Panel Washing & 24/7 Cloud Telemetry

Performance Ratio (PR)	Automated verification against on-site pyranometers. Weather-corrected PR audited quarterly.	**Guaranteed PR ≥ 80.0%** Liquidated damages for shortfalls.
Quarterly Drone Thermography	Autonomous aerial radiometric thermal imaging scans all panels to detect micro-cracks and hotspots.	Full thermographic audit report delivered to client management quarterly.
Bi-Monthly Panel Washing	Scheduled cleaning using low-TDS demineralized water and soft brushes during dawn/dusk hours.	Maintains panel soiling losses strictly below 2.5% in dry winter months.
24/7 Telemetry & Rapid Dispatch	Ariel-SaaS™ Cloud Gateway monitors every string. Faults trigger automated technician dispatch.	**4-Hour Emergency Response SLA** in Gazipur, Narayanganj, Savar & Chattogram.

NEXT STEPS: COMPLIMENTARY 3D DRONE ROOF FEASIBILITY AUDIT

Zero Cost, Non-Binding Executive Consent & Customized 4-Page Feasibility Report

What We Deliver in Your Free Preliminary Audit:

• **3D Drone Aerial LiDAR Mapping:** High-resolution photogrammetry maps usable roof space, skylights, vents, and shadow obstacles.

• **Structural Load Verification:** Certified civil engineers inspect PEB rafters, purlins, sheet gauges, or RCC slab rebar.

• **Substation Load Audit:** Check transformer kVA, switchgear, and Net-Meter tie-in point.

• **Bankable Deliverable within 5 Working Days:** Customized 4-Page Feasibility Dossier with PVsyst 3D layout, guaranteed generation, and bank financing advisory.

Executive Contact & Audit Booking:
Syed Akramul Hassan — Managing Director & CEO
Ariel Renewable Energy Limited | ArielMC Enterprise Eco-System
Direct Hotline / WhatsApp: +880 1819 273458 | Email: energy@arielmc.com | energy.arielmc.com

